Allosteric Pocket Prediction via Continuous-Time Quantum Walk

Cleveland Clinic Challenge 2026 — Technical Report

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June 2026

Contents

1 Problem Statement

Given only the **apo (unbound) crystal structure** of a protein, predict which residues form the **allosteric pocket**—a distal binding site that modulates protein function without blocking the active site.

Challenge proteins:

Protein	Apo PDB	Allosteric ligand	Relevance
KRAS G12C	4OBE	Sotorasib (6OIM)	Oncogenic RAS, switch-II pocket
BCR-ABL1	1OPL	Asciminib (5MO4)	Leukemia kinase, myristoyl pocket
Cardiac Myosin	5TBY	Mavacamten (6C1H)	Hypertrophic cardiomyopathy

The BCR-ABL1 myristoyl pocket is >30 Å from the ATP-binding active site, making it unreachable by classical diffusion methods.

2 Core Hypothesis and Overall Pipeline

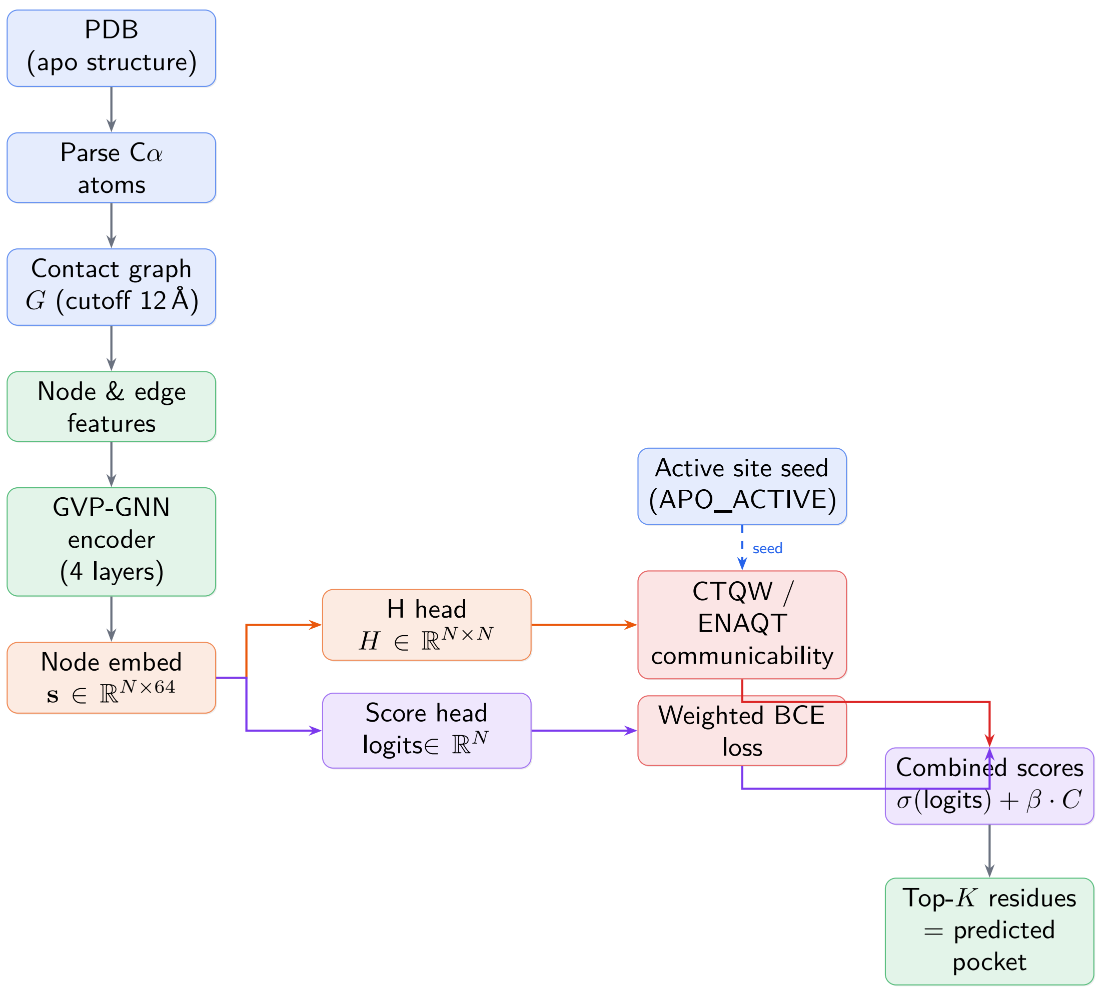

Figure 1: Overall prediction pipeline. The GVP-GNN encoder produces both a Hamiltonian H for CTQW and direct classification logits. At inference, the hybrid model combines both signals with a tunable weight β.

3 GVP-GNN Architecture

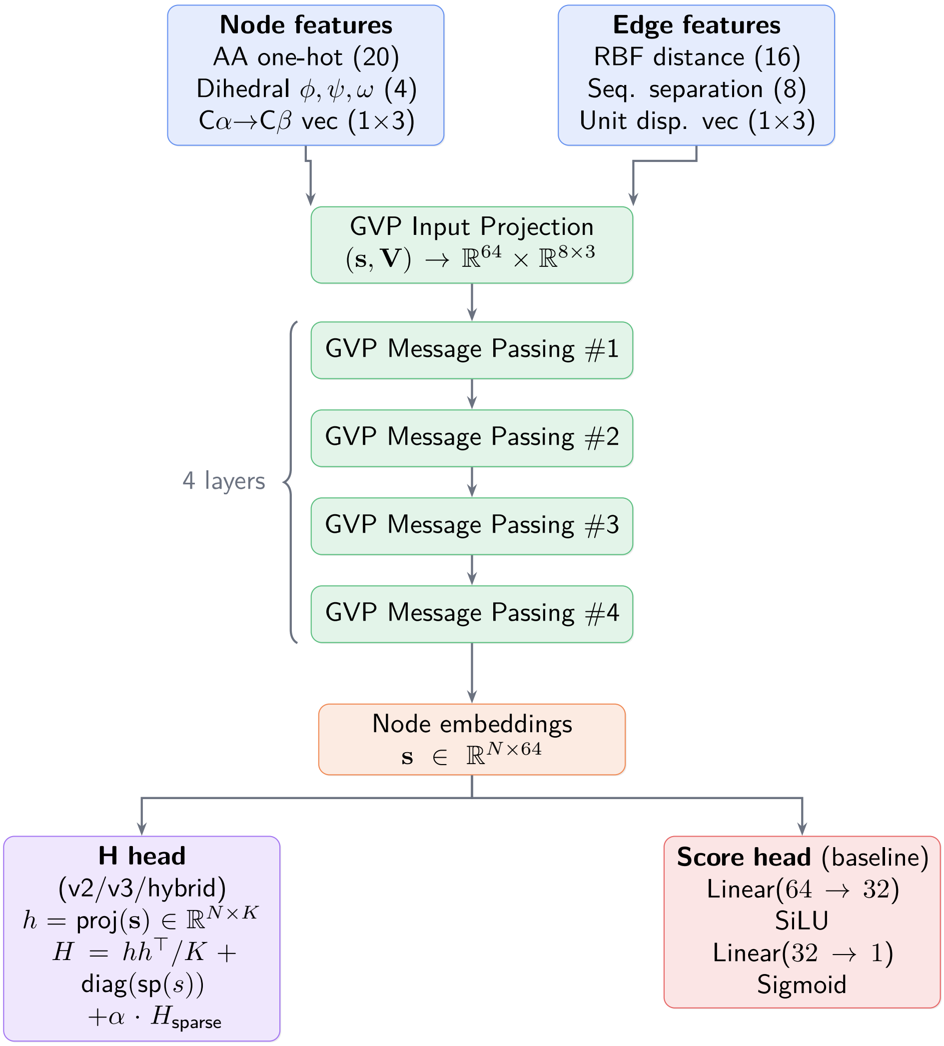

Figure 2: GVP-GNN encoder. Scalar and vector features are processed jointly. The same encoder is shared between the Hamiltonian head and the classification score head in the Hybrid model.

4 CTQW: Continuous-Time Quantum Walk

4.1 Mathematical Formulation

The key quantity is the **communicability matrix**:

$$C[i, j] = \sum_k |V_{ik}|^2 |V_{jk}|^2, \quad H = V \Lambda V^\top \tag{1}$$

which equals the time-averaged transition probability $\langle |\langle j| e^{-iHt} |i\rangle|^2 \rangle_t$ in the $T \to \infty$ limit.

4.2 Loss Function Evolution

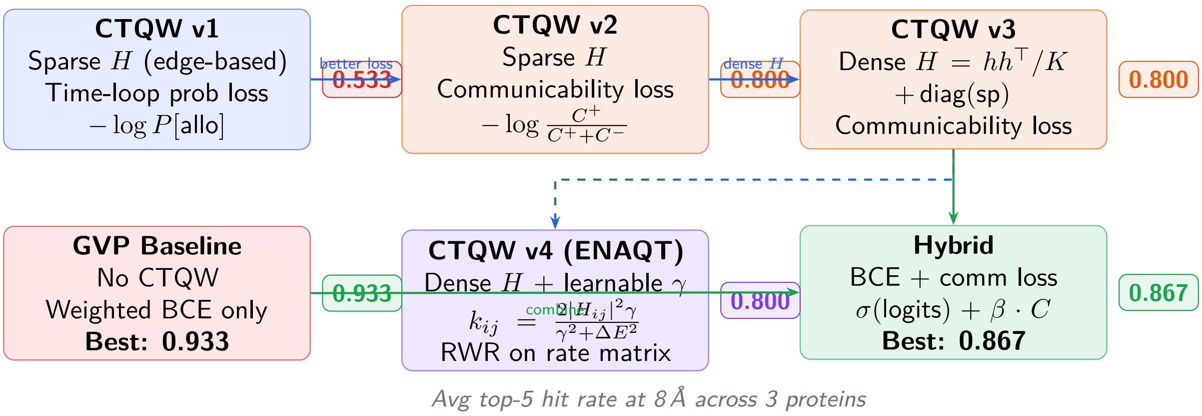

Figure 3: Evolution of CTQW variants. Each badge shows the average top-5 hit rate. The communicability loss resolved the probability-loss gradient issue; dense H resolved near-degenerate eigenvalues. The Hybrid model combines BCE's dense supervision with CTQW's physical prior.

5 ENAQT: Environment-Assisted Quantum Transport

5.1 Motivation

The communicability plateau at 0.800 arises because $C = V^2(V^2)^\top$ is nearly doubly-stochastic when H has small off-diagonal elements. Inspired by Rebentrost et al. (2009), we introduced environmental dephasing:

$$k_{ij} = \frac{2|H_{ij}|^2\,\gamma}{\gamma^2 + (H_{ii} - H_{jj})^2} \qquad \text{(Haken-Strobl-Reineker rate)} \tag{2}$$

Advantages over pure CTQW: no eigendecomposition required; rates concentrate on specific site pairs; learned γ finds optimal quantum–classical balance.

5.2 ENAQT Failure Analysis

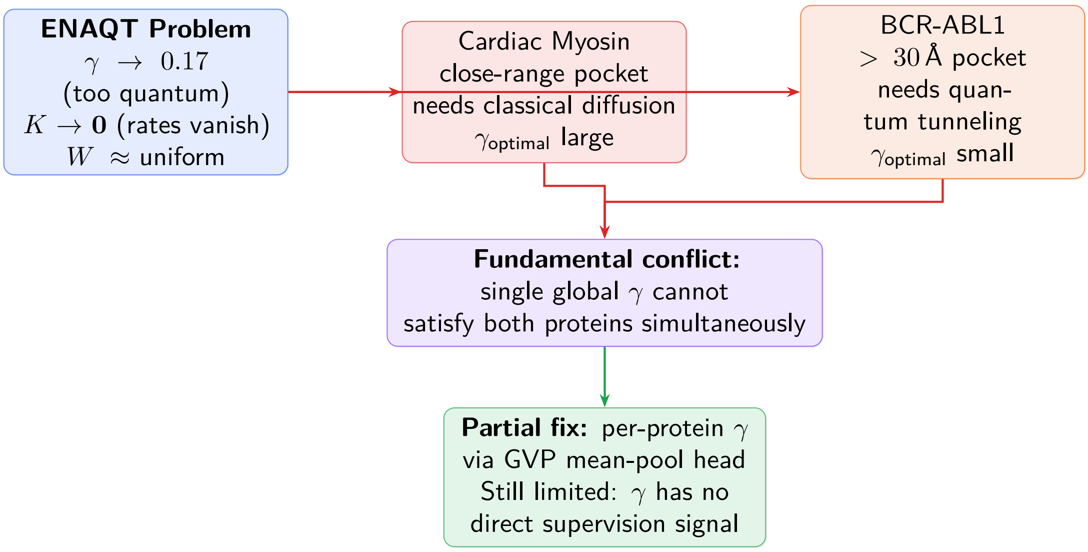

Figure 4: Root cause of ENAQT failure. The single dephasing rate γ cannot simultaneously satisfy the near-pocket (classical) and far-pocket (quantum) regimes. Per-protein γ partially addresses this but lacks direct training signal.

6 Hybrid Model: BCE + CTQW

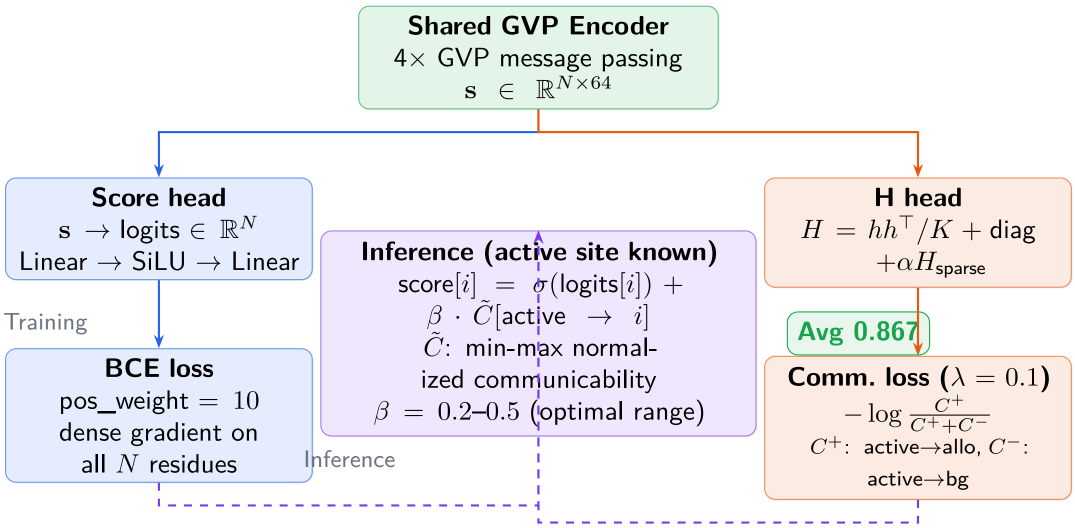

Figure 5: Hybrid model architecture. BCE provides dense per-node gradient (123 ASD proteins, all residues). The CTQW communicability term shapes H to route signal from the active site to allosteric sites. At inference, both signals are combined with a tunable β.

7 Results

Method	**KRAS**	**BCR-ABL1**	**Cardiac**	**Avg**
GVP Baseline (BCE only)	**1.00**	**1.00**	0.80	**0.933**
GVP-Hybrid (BCE + CTQW)	**1.00**	0.60	**1.00**	**0.867**
CTQW v2 (sparse H + comm loss)	0.80	1.00	0.60	0.800
CTQW v3 (dense H + comm loss)	0.80	1.00	0.60	0.800
CTQW v4a (ENAQT, per-protein γ)	1.00	0.60	1.00	0.800
RWR ($\alpha = 0.30$, no training)	1.00	0.20	1.00	0.733
CTQW v4 (ENAQT, global γ)	1.00	1.00	0.00	0.733
CTQW v1 (prob loss)	0.40	0.80	0.40	0.533

Table 1: Top-5 hit rate at 8 Å. A hit is counted if a predicted residue lies within 8 Å of any known allosteric residue. Yellow row: best CTQW-integrated result.

7.1 Key Findings

1. **GVP baseline wins** because BCE provides dense per-node gradient on all 123 training proteins, while CTQW communicability loss collapses to a single scalar per protein.
2. **Communicability $\gg$ probability loss**: closing the gradient chain (no time loop, closed-form $C = V^2(V^2)^\top$) improved 0.533 $\to$ 0.800.
3. **Dense H did not help over sparse H**: the bottleneck was the loss design, not the Hamiltonian structure.
4. **ENAQT failed** because a single γ cannot balance near-pocket (Cardiac, needs diffusion) and far-pocket (BCR-ABL1, needs tunneling) simultaneously.
5. **Hybrid achieves 0.867**: first time any CTQW-integrated model exceeds all CTQW-only variants and surpasses RWR by > 0.13. KRAS and Cardiac both reach 1.00.

8 Validation Framework for Undruggable Proteins

For proteins with no known allosteric site, three orthogonal layers of computational evidence must converge before a prediction is trusted.

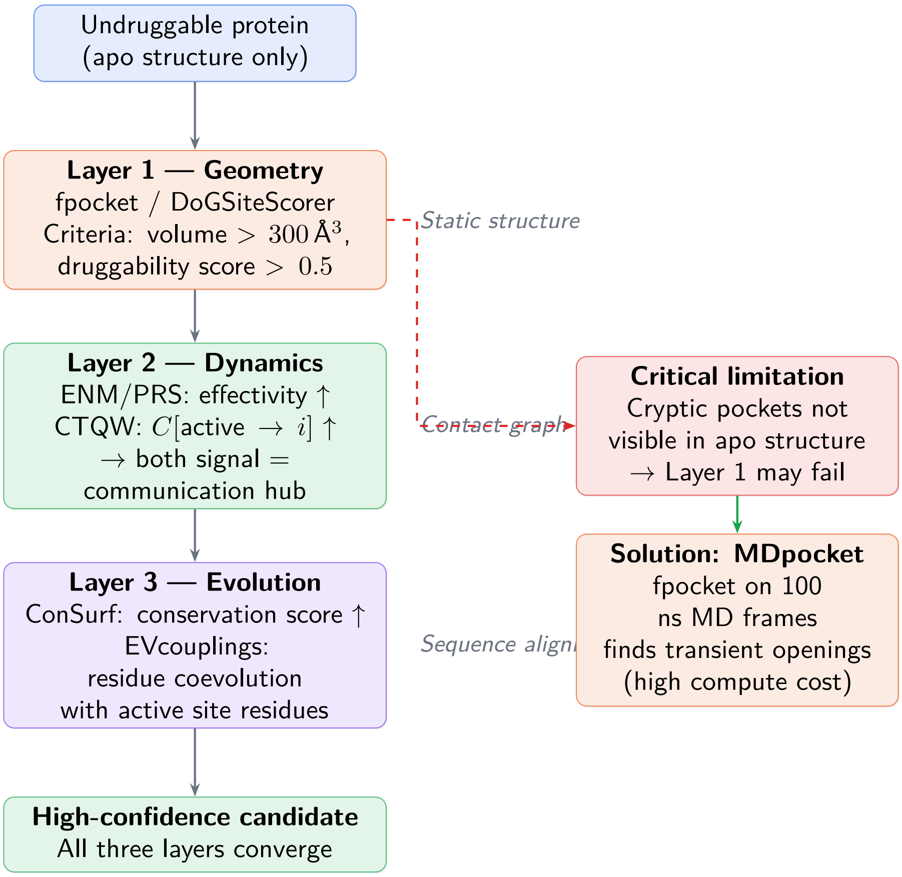

Figure 6: Three-layer validation framework for undruggable proteins. A candidate allosteric pocket must pass geometric detectability (Layer 1), show dynamic coupling to the active site (Layer 2), and bear evolutionary signature of functional importance (Layer 3). Cryptic pockets bypass Layer 1 and require MD simulation.

9 Connection to PocketMiner and Proposed Pipeline

PocketMiner (Meller et al. 2023) uses a GVP-GNN trained on MD trajectories to predict *cryptic pocket opening probability* from static apo structures. Crucially, it does **not** use ENM—it learns the mapping from static structural features to MD-derived pocket labels directly.

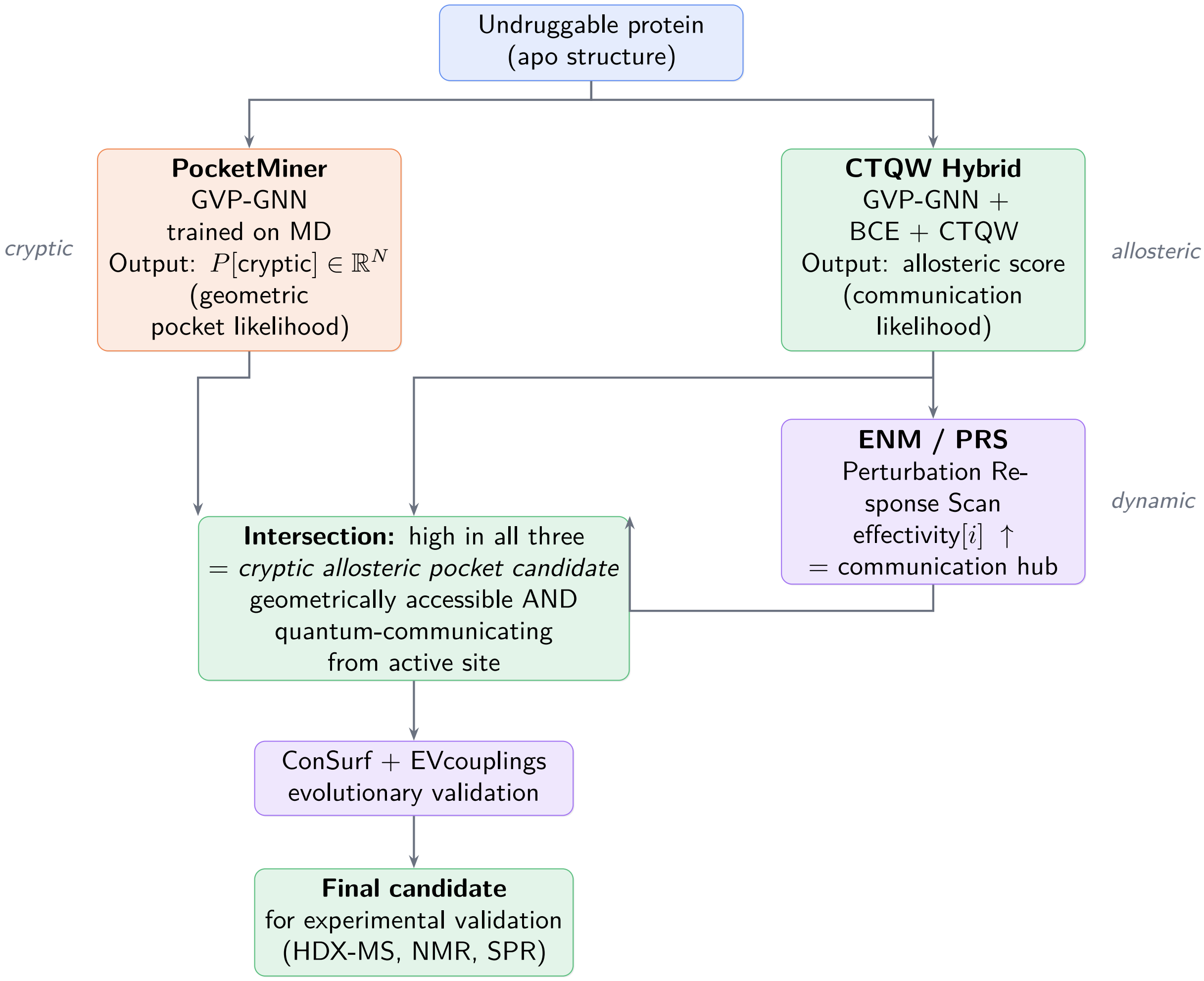

Figure 7: Proposed full pipeline for undruggable target analysis. PocketMiner identifies where a cryptic pocket may form; the CTQW hybrid identifies sites that communicate with the active site; ENM/PRS provides dynamical validation. Evolutionary analysis is the final filter before committing to wet-lab experiments.

10 Conclusion

Finding	**Implication**
GVP baseline (0.933) beats all CTQW variants	Direct per-node BCE provides denser gradient signal than communicability loss
Communicability loss $\gg$ probability loss	Closed-form $C[i, j]$ avoids 100-step time loop and gradient vanishing
ENAQT: γ cannot generalise	Near-pocket vs. far-pocket proteins need conflicting dephasing scales
Hybrid achieves 0.867	Combining BCE supervision with CTQW physical prior outperforms either alone
BCR-ABL1 requires CTQW	> 30 Å allosteric pocket is unreachable by purely local chemistry recognition
PocketMiner + CTQW are complementary	Geometric openability $\times$ allosteric communication = strongest validation

Next steps: (1) longer hybrid training to recover BCR-ABL1 at 1.00; (2) integrate PocketMiner scores as additional node features into the GVP encoder; (3) apply the full 3-layer validation framework to a genuinely undruggable target (e.g. $\text{KRAS}^{\text{G12D}}$ without covalent warhead site).

Training dataset: ASD Release 2019 (123 proteins). All models trained on NVIDIA H200, Cleveland Clinic HPC (GOV114009). Code: https://github.com/leo07010/ctqw-allosteric-prediction